

Mohammed SMIRÈS

SYSTEMS & AI ARCHITECT —  
MBSE, EMBEDDED & COMPLEX  
SYSTEMS



Engineering graduate from ENSEIRB MATMECA with 10 years of industrial experience, specialized in MBSE modeling and requirements definition for complex systems, and recently certified in artificial intelligence at CNAM. I bring to your Systems Architecture or applied AI projects a rigorous systemic approach, proven MBSE practices, and the integration of optimization algorithms to accelerate the design and validation of complex architectures.

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Education

- Certificate of specialization in artificial intelligence  
CNAM, Paris  
From Sept. 2025 to Jun. 2026
- Engineering Degree  
ENSEIRB-MATMECA, Bordeaux  
From Sept. 2010 to Sept. 2014  
Engineering degree in Electronics/Control systems
- Diplôme de Bachelor  
Federal University of Rio Grande do Sul, Porto Alegre, RS, Brazil  
From Jan. 2011 to Jun. 2011  
University exchange (BRAFITEC program)

Work experience

- Systems Engineer  
Thales, Lisbonne/Cholet  
From Feb. 2023 to Mar. 2026  
**Systems Engineering & Requirements:**
  - Specification & Lifecycle Management: Definition, structuring, and monitoring of system requirements to ensure compliance with operational needs.**Architecture & Modeling:**
  - MBSE Design: Structuring system architectures according to the MBSE (Model-Based Systems Engineering) methodology, ensuring overall model consistency and traceability.**Configuration & Range Management:**
  - Variability & Documentary Reference: Analysis of product variability and reorganization of technical documentation to streamline the management of variations.**Performance & R&D:**
  - Technical Performance Management: Management, feasibility analysis, and development for the performance optimization of new designs.
  - Facilitation of international collaborations, with interventions at the Thales site in Hengelo (Netherlands).
- Embedded Systems Engineer  
Critical software, Lisbon  
From Aug. 2021 to Feb. 2023  
**Rolling Stock Project (Oxygène Trains – SNCF / CAF):**
  - Systems Engineering & Specifications: Definition of system requirements based on functional specifications (signaling, driving modes, coupling, radio).
  - Tooling & Python Automation: Development of a Python testing tool automating the reading and execution of procedures.
  - Deployments & Site Collaboration: Regular interventions and travel to the manufacturer's site (CAF – Beasain, Basque Country) for monitoring and technical alignment.**C++ Validation & Tests:**
  - Automatic Train Protection (ATP): Implementation of a C++ test library for the ATP system and drafting of associated validation procedures.
- Test and Validation Engineer  
Inetum, Lisbonne  
From Dec. 2020 to Aug. 2021  
**Software Validation & Quality:**
  - Web Testing & Qualification: Elaboration and execution of test plans for the functional validation of Web applications and their interfaces (HMI).**Technical & Data Support:**
  - Database Analysis & Resolution: Diagnosis, investigation, and correction of malfunctions related to database servers.

## Skills

### Engineering & MBSE

Arcadia, Capella, Requirements Management & Traceability, Component-Oriented Architecture.

### AI & Data

Artificial Intelligence, Optimization algorithms.

### Model Based Design (MBD)

MATLAB / Simulink, Systems Modeling & Simulation.

### Development & Testing

C/C++, Python, Jenkins, Test automation.

### Systems requirements management

Doors, Polarion.

## Languages

### French

Bilingual / Native

### English

Fluent (Professional)

### Portuguese

Advanced

## Travel

### New Zealand

Road-trip in 2020

## Social networks

 [mosmires.github.io](https://github.com/mosmires)

 <https://www.linkedin.com/in/smires/>

## Work experience

### Control Systems Engineer | Engine Air Flow Regulation Manager

From Jun. 2018 to Oct. 2019

Bertrandt, Toulouse

#### *Algorithms & Engine Control:*

- **Algorithm Development (Flex Fuel):** Design of an ethanol rate estimation algorithm via data fusion from tank/injector sensors.

#### *Embedded Software Engineering & Core Component:*

- **Component Manager (Core - Air Management):** Fault diagnosis, anomaly detection, and resolution via Software patches, Redesign, or Calibration.
- **Simulink Modeling & Automation:** Development of a tool for the automatic plotting of multi-core architecture on Simulink directly from the embedded software.

#### *Tooling & Validation:*

- **Processing Tooling:** Creation of internal tools (test bench and vehicle) for test data analysis and multi-core bug detection.

### Engine Control Design and Validation Engineer

From Mar. 2016 to Jan. 2018

Assystem, Paris

#### *Engine Oil Algorithms & Functions:*

- **Innovative Strategy & Mass Calculation:** Design of an advanced logic for estimating oil mass [Engine/Crankcase] by fusing data from an ultrasonic sensor.
- **Vehicle Life Management & Alerts:** Definition of oil change management rules and alert thresholds (MIN/MAX levels) adapted to different vehicle life cycles.

#### *Engine Control Validation & Integration:*

- **Validation Strategies & Test Plans:** Elaboration and drafting of integration and validation plans for all engine control strategies.
- **HIL Testing & Prototype Vehicles:** Functional validation of Simulink models and requirements on HIL (Hardware-In-the-Loop) benches and prototype vehicles.

### Calibration Engineer

From Feb. 2015 to Mar. 2016

Set-way, Paris

#### *Engine Calibration & Tuning:*

- **Calibration & Optimization (Gasoline):** Tuning and optimization of engine and vehicle parameters on Gasoline motorization projects.
- **Methodology Drafting:** Elaboration of structured calibration methodologies, based on the Renault repository and specifications.

### End of Studies Internship

From Mar. 2014 to

SKF, Vernon

Sept. 2014

#### *Advanced Automatics & Mechatronics:*

- **Robust Control (CRONE) for vibration isolation:** Design and synthesis of a non-integer order control law for damping the rigid and flexible modes of a rotor on an active magnetic bearing.